Portfolio Optimization III

This document shows the forecasted paths for each stock for June using Monte Carlo and how each stock has performed.

Linear Programming Model

In order to find the optimal number of shares for each stock that maximises the returns on 29 June 2026 starting on 31 May 2026, a linear programming model is used run in Microsoft Excel using solver. The model is presented below.

Let X_i be the number of shares of stock i bought, where $i \in \{1, 2, 3, 4\}$

$1 \rightarrow$ Apple
$2 \rightarrow$ Microsoft
$3 \rightarrow$ JP Morgan
$4 \rightarrow$ Exxon

Maximize $322.31X_1 + 457.49X_2 + 310.15X_3 + 149.19X_4$ (Forecasted close prices at 29 June)

Subject to:

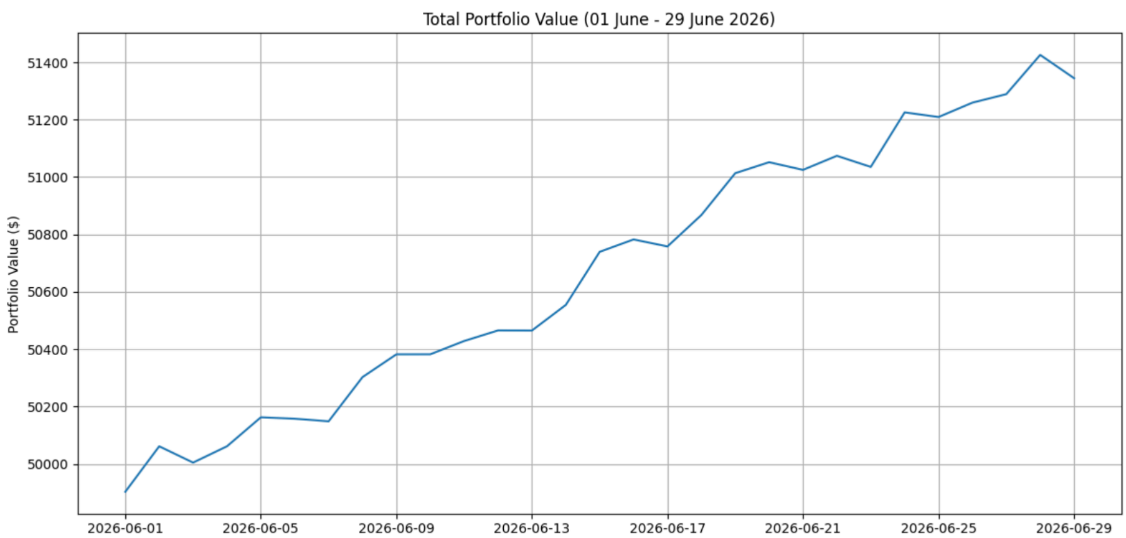
$$
\begin{aligned}
312.59X_1 + 450.72X_2 + 299.82X_3 + 145.46X_4 &\leq 50\,000 \quad \text{(Total available funds are \$50 000)}\\
312.59X_1 + 299.82X_3 &\geq 25\,000\\
450.72X_2 + 145.46X_4 &\geq 20\,000\\
299.82X_3 + 145.46X_4 &\geq 15\,000\\
310.15X_3 &\leq 15\,000
\end{aligned}
$$

$$
\begin{aligned}
X_3 &\leq 80 \quad \text{(JP Morgan shares less than 80)}\\
X_4 &\leq 100 \quad \text{(Exxon shares less than 100)}\\
X_i &\geq 10 \quad \forall\, i \quad \text{(Atleast 10 shares for each stock)}
\end{aligned}
$$

The optimal number of shares to be purchased and the associated value of TP are as follows:

X_1 = 19.24 **(19)**; X_2 = 12.10 **(12)** ; X_3 = **80** and X_4 = **100**

Starting Point = **$49 879.45**
Total Portfolio Value = **$51 344.77**
Portfolio Commission = **$1 465.32**
Effective Return = **2.94%**



Portfolio Analysis

This section analyses the expected returns and total variance-covariance and of the portfolio.

- **Portfolio Expected Return**

$$E(R_p) = \sum_{i=1}^{n} w_i E(R_i)$$

Stock	Shares	Opening Share Price	Forecasted Share Price	**Weights (W_i)**	**Expected Returns (E_i)**	**$W_i\, E(R_i)$**
Apple	19	312.59	322.31	**11.91**	**3.11**	**0.3704**
Microsoft	12	450.72	457.49	**10.84**	**1.50**	**0.1626**
JP Morgan	80	299.82	310.15	**48.09**	**3.33**	**1.6014**
Exxon	100	145.46	149.19	**29.16**	**2.50**	**0.7290**
Total	211	1208.59	1239.14	**100**	**10.44**	**2.8634**

The expected returns of this portfolio is **2.86%** as shown which is consistent with the effective return calculated above in the linear programming model above.

- **Portfolio Variance**

$$\sigma_p^2 = \mathbf{w}^T \mathbf{\Sigma} \mathbf{w}$$

The matrices below are the weights (***w***) and the variance-covariance matrix (**Σ**). The weights-transpose (***w***^{T}) can be obtained through transposing the weights matrix

$$\begin{bmatrix} 0.1191 & 0.1084 & 0.4809 & 0.2916 \end{bmatrix}$$

$$\begin{bmatrix} 287.10 & 52.09 & 40.36 & 19.78 \\ 52.09 & 934.27 & 184.36 & -307.19 \\ 40.36 & 184.36 & 128.21 & -80.31 \\ 19.78 & -307.19 & -80.31 & 132.28 \end{bmatrix}$$

Carrying out the multiplication shows that the total portfolio variance is **40.568** (standard deviation of 6.369). Using the variance-covariance matrix (shown below) which includes the forecasted values gives a variance of 54.564 which is standard deviation of 7.387

$$\Sigma = \begin{bmatrix} 660.49 & 402.52 & 47.67 & 14.99 \\ 402.52 & 1013.25 & 154.68 & -235.24 \\ 47.67 & 154.68 & 101.29 & -60.49 \\ 14.99 & -235.24 & -60.49 & 101.59 \end{bmatrix}$$

Conclusion and Recommendations

This portfolio consists of 211 shares; 19 Apple Inc, 12 Microsoft Corporation, 80 JP Morgan Chase & Co and 100 Exxon Mobil Corporation bought at the closing price on 31 May 2026. Using the 50th percentile projections from a Monte Carlo Simulation, the portfolio has 2.86% expected returns and total variance of $40.568. The forecasts are only for 29 days, from 01 June to 29 June 2026. In money terms $49 879.45 was invested and is expected to grow to $51 344.77 over 29 days, which is a commission of $1 465.32 in just a month.

The limitations are; the portfolio has little diversification, hence each stock makes up more that 10% which opens the portfolio up to loss should the stocks drop drastically. The portfolio is missing scenario analysis in that it doesn’t explicitly discuss best case and worst case scenarios and only looks at the median performance. The portfolio also only uses one model to forecast the stock prices (Monte Carlo Simulation) which does not provide model diagnostics and greatly reduces daily returns, misrepresenting the volatility expected in real-life situations.

Overall, 2.86% is a significant increase in portfolio value in one month especially noting that the portfolio has only 5 stocks that are highly volatile. It is advisable to do a scenario analysis to look at how the portfolio would perform under extreme market conditions and to add more stocks to the portfolio.
